Animal Identification in the Wild

A case study in practical data science

Context

You are a junior forest ranger in the U.S. Wildlife Service. You and your colleagues have just spent the last couple of years installing camera traps in wildlife locations all across America in order to better document and understand animal behaviors, habitats, and migration patterns. Camera traps are camouflaged cameras that take pictures whenever movement occurs.

A camera trap attached to a tree trunk.

Motivation

Camera traps take pictures when there is movement in the frame - any movement. You and your colleagues are being overwhelmed with pictures with no animals in them! Being the bright young tech-savvy person that you are, you remember that you can use Convolutional Neural Networks (CNNs) to detect when an animal is in an image. Not only that, but you know for certain that you can also train a CNN model to detect what species of animal is in the image.

Your boss, a senior ranger, is overjoyed at hearing the news, and is delighted at the idea of not only being able to get rid of false positives, but of also being able to filter images for specific animal species. No more sorting through images by hand! He not only green-lights your project idea, but also gives you access to a High Performance Computing (HPC) cluster with GPUs.

Topic

Your goal is to use data science and machine learning to create a CNN that can scan images and detect what species of animal is in them, or if they're empty, with at least 95% accuracy.

General Process

In this case study, you will have to (1) select and download training data, (2) research and select existing CNN model architecture that suits your goal, (3) write code to train and test the model, and (4) execute that code on the HPC.